NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Hand & Upper Limb

1. flexor and extensor tendons
2. nerve compression and injury
3. brachial plexus

Flexor and extensor tendons

1. Flexor tendon anatomy

- **FDS** splits at the proximal phalanx (**Camper's chiasma**) and inserts on the **middle phalanx**; **FDP** passes through the split to the **distal phalanx base**. **FDP of index acts independently** (separate muscle belly); middle/ring/little FDP share a common belly (**quadriga effect**).
- **FDS to the little finger is absent or weak in about 15-20%**.
- **Flexor sheath pulleys: five annular (A1-A5)** and **three cruciform (C1-C3)**; **A1, A3, A5 over joints (MCPJ, PIPJ, DIPJ)**; **A2 (proximal phalanx) and A4 (middle phalanx)** are **the most important biomechanically** (prevent **bowstringing**). **Thumb:** A1, **oblique pulley (most important)**, A2.
- **Nutrition: synovial diffusion** (major within the sheath) and **vincula** (**vinculum longum and breve**, carry blood via transverse digital arteries); **watershed avascular zones** on the volar aspect of the tendons.

2. Flexor tendon zones (Verdan/Kleinert)

Zone	Extent	Notes
I	Distal to FDS insertion (FDP only)	**Jersey finger** (FDP avulsion)
II	**A1 pulley (distal palmar crease) to FDS insertion**	**"No man's land"** (Bunnell): **both FDS and FDP within the tight fibro-osseous sheath**; adhesions; historically poor results
III	Distal edge of carpal tunnel to A1 pulley	**Lumbrical origins** (palm); better prognosis
IV	**Carpal tunnel**	Median nerve injury associated
V	Proximal to carpal tunnel (forearm)	Multiple tendons, nerves and arteries ("**spaghetti wrist**")
Thumb T I-T V	T I distal to IPJ; **T II A1 to IPJ**; T III thenar eminence; T IV carpal tunnel; T V forearm	FPL

3. Flexor tendon repair

- **Timing: primary repair (within 24 hours)** or **delayed primary (within about 10-14 days)**; beyond **3-4 weeks**: tendon retraction and myostatic contracture: consider **staged reconstruction**.
- **Principles:** atraumatic handling, **multi-strand core suture** (**strength is proportional to the number of strands crossing the repair site**: **4-strand or 6-strand core** (e.g. **Adelaide, cruciate, modified Kessler/Strickland**) allows **early active motion**), **core purchase about 7-10 mm from the cut end**, **dorsally placed core suture** is stronger, **epitendinous running suture** (adds about 10-50% strength and reduces gap/gliding resistance), **locking loops** add strength.
- **Gap formation over 3 mm** at the repair: increased **rupture and adhesions**.
- **Repair strength:** decreases between **days 5 and 21** (softening phase) and recovers after that.
- **Pulleys:** preserve **A2 and A4**; **venting (partial release) of A2/A4 (up to about two-thirds of A2 or the whole A4 if needed)** allows passage of bulky repairs without significant bowstringing.
- **FDS in zone II:** repair both slips if possible; **repair only FDP (excise one slip of FDS)** when the canal is too tight.
- **Rehabilitation:**
  - **Early active motion** (**Saint John protocol, Manchester short splint**, true active flexion with 4+ strand repairs): **best range of motion**; slightly higher rupture than passive regimens in some studies.
  - **Kleinert (active extension, passive flexion with rubber bands)**, **Duran (controlled passive motion)**.
  - **Dorsal blocking splint**: **wrist neutral to slight flexion (0-20 degrees), MCPJ flexed about 50-70 degrees, IPJs extended**, for about **4-6 weeks**.
  - Strengthening after about 8-12 weeks.
- **Complications: adhesions** (commonest; tenolysis after **3-6 months** once soft tissues settle and passive ROM is full), **rupture (about 4-10%, usually days 7-14; re-repair if early)**, **joint contracture (PIPJ flexion)**,

bowstringing, **trigger finger**, **quadriga** (over-advanced FDP repair restricts the other fingers' flexion), **lumbrical-plus finger** (paradoxical PIPJ extension on attempted flexion when FDP is cut distal to the lumbrical origin or the graft is too long: lumbrical pulls the extensor apparatus).

4. Jersey finger (FDP avulsion; zone I)

- **Ring finger** commonest (about 75%): forced extension of an actively flexing finger (grabbing a jersey).
- **Leddy and Packer classification:**
  - **Type I:** tendon retracts into the **palm** (both vincula ruptured): **repair within 7-10 days** (poor vascularity, rapid myostatic contracture).
  - **Type II:** retracts to the **PIPJ** (long vinculum intact): repair within about **6 weeks** (commonest type).
  - **Type III: large bony fragment caught at the A4 pulley** (retracts to DIPJ): **ORIF**.
  - **Type IV:** bony fragment + tendon avulsed from the fragment: fix both.
  - **(Type V: with distal phalanx fracture.)**
- **Repair:** pull-out button, **suture anchors**.
- **Late presentation:** DIPJ arthrodesis/tenodesis, accept the deficit (FDS still flexes the PIPJ), **two-stage grafting** (rarely).

5. Flexor tendon reconstruction

- **Single-stage free tendon graft** (palmaris longus: absent in about 15%; plantaris; toe extensors) if the bed is good and pulleys intact.
- **Two-stage (Hunter) reconstruction:** stage 1: **silicone rod** inserted to form a **pseudosheath** (and pulley reconstruction), **stage 2 (about 3 months)**: tendon graft through the new sheath. For **scarred beds, damaged pulleys**.
- **Boyes preoperative classification:** grade 1 good (minimal scar) to grade 5 salvage (multiple digits, poor prognosis).
- **Pulley reconstruction:** tendon graft looped around the phalanx (**triple loop for A2**), **under the extensor mechanism for A2, over it for A4** (around the bone).

6. Extensor tendon anatomy

- **Six dorsal wrist compartments (radial to ulnar):**
  - **APL + EPB** (**De Quervain's tenosynovitis**; **separate EPB subsheath** in many: release it too),
  - **ECRL + ECRB** (**intersection syndrome**: proximal, where compartment 1 muscles cross compartment 2: about **4-8 cm proximal to Lister's tubercle**),
  - **EPL** (ulnar to **Lister's tubercle**: **attrition rupture after undisplaced distal radius fractures**, typically 4-8 weeks later),
  - **EDC + EIP** (**posterior interosseous nerve** on the floor),
  - **EDM** (overlies the DRUJ),
  - **ECU** (subsheath; ECU subluxation).
- **Juncturae tendinum** (between EDC tendons, dorsum of the hand): mask proximal EDC lacerations (finger can still extend).
- **EIP and EDM lie ulnar to the EDC tendons** of their digit (independent extension); **EIP used for EPL reconstruction** (tendon transfer).
- **Extensor mechanism of the finger:**
  - **Central slip** (EDC) inserts on the **base of the middle phalanx** (extends the PIPJ),
  - **Lateral bands** (from **interossei and lumbricals** plus EDC): join to form the **terminal tendon** at the distal phalanx (extends DIPJ),
  - **Sagittal bands** (centralise the EDC over the MCPJ; extend the MCPJ via a sling on the proximal phalanx),
  - **Triangular ligament** (dorsally, between the lateral bands: prevents volar subluxation),
  - **Transverse retinacular ligament** (prevents dorsal subluxation of lateral bands),
  - **Oblique retinacular ligament (Landsmeer):** volar to the PIPJ axis, dorsal to DIPJ: **links PIPJ and DIPJ extension** (tight in boutonniere deformity).
- **Intrinsics: flex the MCPJ and extend the IPJs** (lumbricals originate from FDP: **the only muscles that relax their own antagonist**).

7. Extensor tendon zones (Verdan)

Zone	Location	Classic injury
I	**DIPJ**	**Mallet finger**
II	Middle phalanx	Lateral band / terminal tendon lacerations
III	**PIPJ**	**Boutonniere (central slip)**
IV	Proximal phalanx	Partial lacerations (broad tendon)
V	**MCPJ**	**Fight bite (human bite)**; **sagittal band rupture** ("**boxer's knuckle**")
VI	Metacarpal	Lacerations proximal to the juncturae
VII	**Wrist (extensor retinaculum)**	Retinacular adhesions
VIII	Distal forearm	Musculotendinous junction
IX	Proximal forearm (muscle bellies)	

- **Odd zones over joints** (I, III, V, VII); thumb T I (IPJ) to T V.

8. Mallet finger (zone I)

- **Forced flexion of an extended DIPJ**: terminal tendon rupture or bony avulsion of the dorsal base of the distal phalanx. **Untreated: swan-neck deformity** (extensor imbalance: PIPJ hyperextension).
- **Doyle classification:**
  - **Type I:** closed, ± small avulsion fracture (commonest),
  - **Type II:** laceration at/proximal to the DIPJ,
  - **Type III:** deep abrasion with loss of skin and tendon,
  - **Type IV:** mallet fracture: **A** transepiphyseal (children: **Seymour fracture** if open with nail bed injury: **requires irrigation, debridement, nail-bed repair, antibiotics**: risk of osteomyelitis); **B** hyperflexion injury with fracture **20-50% of the articular surface**; **C** hyperextension with **over 50% articular surface** and **volar subluxation of the distal phalanx**.

- **Treatment:**
  - **Most closed mallet injuries (tendinous or bony): continuous extension splinting of the DIPJ (slight hyperextension) for 6-8 weeks full-time**, then night splint for 2-4 weeks (**PIPJ left free**); **any flexion during this time restarts the clock**. Effective even if presentation is delayed (up to about 3 months).
  - **Surgery (relative): volar subluxation of the distal phalanx** and/or **large fragment (over about 30-50% of the articular surface)**: **extension block pinning (Ishiguro)**, ORIF. Open injuries: repair.
  - **Chronic with swan neck: central slip tenotomy (Fowler)**, **spiral oblique retinacular ligament reconstruction (SORL)**, DIPJ arthrodesis.

9. Boutonniere deformity (zone III)

- **Central slip rupture**: PIPJ flexion; lateral bands migrate **volar to the PIPJ axis** (triangular ligament attenuates): become **PIPJ flexors** and **hyperextend the DIPJ**.
- **Acute injury: Elson test** (**most reliable**: PIPJ flexed 90 degrees over the table edge; attempt to extend the middle phalanx against resistance: **with central slip rupture, the DIPJ goes rigid/extends** (lateral bands act); with an intact slip, the DIPJ remains floppy).
- **Treatment:**
  - **Closed acute: PIPJ extension splinting for 6 weeks, DIPJ free (active DIPJ flexion exercises to pull the lateral bands dorsally)**.
  - **Open laceration/large avulsion fracture:** repair/fix.
  - **Chronic: supple: splinting/serial casting first** (most improve); surgical reconstruction (**Fowler/Dolphin terminal tendon tenotomy** for DIPJ hyperextension; central slip reconstruction) only once **passive PIPJ extension is restored**. **Pseudoboutonniere** = PIPJ hyperextension injury with volar plate scarring (flexion contracture without DIPJ hyperextension).

10. Sagittal band rupture ("boxer's knuckle", zone V)

- Usually **radial sagittal band** (middle finger commonest): **ulnar subluxation of EDC** with MCPJ flexion; extension lag at the MCPJ.
- **Acute: sagittal band (relative motion/yoke) splint** for about 6 weeks; **chronic/failed:** surgical repair or **centralisation (sagittal band reconstruction)**.

11. Fight bite (clenched-fist injury, zone V)

- Small wound over the MCPJ: **assume joint penetration and tendon injury**; skin, tendon and capsule wounds move apart when the fingers are extended.
- **Organisms: mixed: *Streptococcus*, *S. aureus*, *Eikenella corrodens* (classic), anaerobes**: **co-amoxiclav (amoxicillin-clavulanate)**; *Eikenella* is resistant to first-generation cephalosporins/clindamycin.
- **Exploration, irrigation and debridement**, leave wound open, splint; tendon repair delayed if contaminated.

12. Other extensor problems

- **EPL rupture after distal radius fracture** (often undisplaced): **EIP to EPL transfer** (commonest) or free graft.
- **Vaughan-Jackson syndrome** (rheumatoid arthritis): **attrition rupture of the ulnar extensor tendons (EDM, then EDC to little and ring) over a dorsally prominent ulnar head (caput ulnae)**: **tenosynovectomy, Darrach/distal ulna procedure, tendon transfers (EIP or side-to-side)**. **Mannerfelt-Norman lesion**: **FPL attrition rupture over a scaphoid osteophyte** (rheumatoid).
- **Extensor repair: zones IV-VIII**: core sutures (thinner tendons in zones I-IV need running/horizontal mattress or figure-of-eight sutures); **relative motion extension splinting (yoke)** for zones IV-VII allows early active motion; **static or dynamic splinting** otherwise.

Exam pearl: **Zone II (A1 pulley to FDS insertion) = no man's land. A2 and A4 pulleys prevent bowstringing; thumb oblique pulley most important. Repair strength depends on the number of core strands; 4-6 strand + epitendinous allows early active motion. Weakest days 5-21; ruptures usually days 7-14. Adhesions = commonest complication.**

Exam pearl: **Jersey finger: ring finger; Leddy-Packer I (palm, both vincula) repair within 7-10 days; III bony fragment caught at A4. Lumbrical-plus finger: paradoxical PIPJ extension. Quadriga: over-advanced FDP.**

Exam pearl: **Mallet: 6-8 weeks continuous DIPJ extension splint (PIPJ free); fix if volar subluxation. Boutonniere: Elson test; PIPJ splinted in extension, DIPJ free to flex. Fight bite: Eikenella, co-amoxiclav, explore. EPL rupture after undisplaced distal radius fracture: EIP transfer. Vaughan-Jackson = ulnar extensors over the ulnar head (RA); Mannerfelt = FPL over the scaphoid.**

Nerve compression and injury

1. Nerve injury basics

Structure

- **Endoneurium** (around each axon), **perineurium** (around a fascicle: the **blood-nerve barrier** and main tensile strength), **epineurium** (around the whole nerve, plus between fascicles).
- **Myelinated fibres:** saltatory conduction at the nodes of Ranvier; **Schwann cells** (PNS).

Seddon and Sunderland classifications

Seddon	Sunderland	Damage	Tinel sign	Recovery
Neurapraxia	**I**	**Local conduction block (demyelination)**, axon intact	**Absent (no advancing Tinel)**	**Complete, within days to about 12 weeks**
Axonotmesis	**II**	**Axon disrupted, endoneurium intact**	**Advancing**	**Complete** (Wallerian degeneration, regrowth along intact tubes)
	III	Axon + **endoneurium** disrupted, perineurium intact	Advancing	**Incomplete/variable** (mismatching, scarring)
	IV	Axon, endoneurium + **perineurium** disrupted; **epineurium intact** (neuroma-in-continuity)	Static (does not advance)	**None without surgery** (excise and repair/graft)

Seddon	Sunderland	Damage	Tinel sign	Recovery
Neurotmesis	**V**	**Complete transection**	Static	**None without surgery**
	VI (Mackinnon)	**Mixed** pattern		Variable

- **Wallerian degeneration:** the distal segment degenerates (**begins within 24-48 hours**); macrophages and Schwann cells clear debris; Schwann cells form **bands of Büngner** to guide regeneration.
- **Regeneration rate:** about **1 mm/day (about 1 inch/month)** after a latent period; **advancing Tinel sign** tracks it.
- **Motor end plates degenerate irreversibly after about 12-18 months** of denervation: **motor recovery requires reinnervation before this**: hence **nerve transfers** close to the target for proximal injuries. Sensory receptors survive longer.
- **Prognosis is better with: young age**, **distal (closer to the target) injuries**, **pure motor or pure sensory nerves**, **sharp clean injuries**, **early repair**, **short gaps**; **worse with:** old age, proximal injuries, mixed nerves, crush/avulsion, delay, smoking/diabetes.

Investigations

- **Nerve conduction studies (NCS) and EMG:** wait **about 3 weeks** after injury (Wallerian degeneration must occur before denervation changes appear). **Fibrillation potentials and positive sharp waves** = denervation (axonal injury); **motor unit potentials (nascent, polyphasic)** = reinnervation (appear before clinical recovery).
- **Sensory nerve action potentials (SNAPs) preserved with numb skin = preganglionic (root avulsion) lesion** (the dorsal root ganglion cell body is intact) (brachial plexus).
- **Ultrasound/MRI neurography:** continuity, neuroma.

Repair principles

- **Primary repair (sharp injuries)** within days; **tension-free**; **epineurial repair** (most common; align with surface vessels, fascicular pattern),

group fascicular repair where fascicles are topographically defined (e.g. motor branch of the ulnar nerve at the wrist).

- **Gaps: nerve conduit** (hollow tube; **gaps under about 3 cm in small-calibre sensory nerves**), **processed nerve allograft** (gaps up to about 5-7 cm in sensory nerves), **autograft (sural nerve: gold standard for larger gaps)**. Avoid tension (**over about 10% elongation impairs blood flow**).
- **Nerve transfers** (e.g. **AIN to deep motor branch of ulnar nerve** (supercharge end-to-side), **Oberlin transfer** (ulnar fascicle to biceps branch), **spinal accessory to suprascapular**, **radial nerve triceps branch to axillary nerve**): convert a proximal injury into a distal one.
- **Tendon transfers** for late or irrecoverable motor loss.
- **Closed injuries with fractures (e.g. radial nerve with humeral shaft fracture):** most are neurapraxia/axonotmesis: **observe** with serial examination and EMG at about 3 months; **explore** if **no recovery by about 3-6 months**, open fractures, **nerve palsy after manipulation (relative)**, penetrating trauma.

2. Carpal tunnel syndrome (CTS)

- **Commonest compressive neuropathy.** Carpal tunnel boundaries: **floor**: carpal bones; **roof**: **transverse carpal ligament (flexor retinaculum)** from **scaphoid tubercle and trapezium** (radial) to **pisiform and hook of hamate** (ulnar). Contents: **median nerve + 9 flexor tendons (4 FDS, 4 FDP, FPL)**.
- **Associations: female**, **pregnancy**, **diabetes**, **hypothyroidism**, **rheumatoid arthritis**, **obesity**, **acromegaly**, **amyloidosis** (dialysis, **transthyretin amyloid: carpal tunnel often precedes cardiac amyloidosis**), **gout**, mass lesions, **distal radius fracture (acute CTS: urgent release if progressive)**.
- **Symptoms: nocturnal paraesthesia** in the radial three and a half digits (**palm spared**: **palmar cutaneous branch** arises about **5-6 cm proximal to the wrist crease and passes superficial to the retinaculum**), clumsiness, **thenar wasting (APB)** in severe cases.
- **Tests:**
  - **Durkan (carpal compression) test** (**most sensitive**),

  - **Phalen test** (wrist flexion 60 s),
  - Tinel sign,
  - **Semmes-Weinstein monofilament** (**most sensitive threshold test for early compression**), **two-point discrimination** (innervation density: **late** change).
  - **CTS-6** clinical diagnostic tool.
- **NCS:** prolonged **distal sensory latency** (earliest), motor latency (**over 4.5 ms**), reduced amplitudes (axonal loss); **EMG** fibrillations in APB (severe).
- **Treatment:**
  - **Night splinting in neutral**, **corticosteroid injection** (also diagnostic: **good response predicts a good surgical outcome**), treat cause.
  - **Carpal tunnel release (open or endoscopic):** similar long-term results; endoscopic: faster return to work, **higher risk of transient nerve injury**. Release **in line with the radial border of the ring finger** (avoids the **recurrent motor branch** and the **palmar cutaneous branch**; **Kaplan cardinal line**).
  - **Recurrent motor branch variants: extraligamentous recurrent (commonest, about 50%)**, **subligamentous**, **transligamentous (about 25%; at risk)**.
  - **Pillar pain** (commonest complaint after release), scar tenderness, **incomplete release** (commonest cause of persistent symptoms), recurrence, **CRPS**, nerve injury.

3. Pronator syndrome and AIN syndrome

- **Pronator syndrome:** median nerve compression in the **proximal forearm**: **sensory symptoms including the palm (palmar cutaneous branch territory)**, **no nocturnal symptoms**, **NCS usually normal**. **Compression sites (proximal to distal):**
  - **Ligament of Struthers** (from a **supracondylar process** of the humerus, about 5 cm above the medial epicondyle),
  - **Lacertus fibrosus (bicipital aponeurosis)**,
  - **Between the two heads of pronator teres** (**commonest**),
  - **FDS arch (sublimis bridge)**.

  - **Provocative tests:** resisted pronation with elbow extension (pronator teres), resisted elbow flexion with supination (lacertus), resisted middle finger PIPJ flexion (FDS arch).
- **Anterior interosseous nerve (AIN) syndrome: pure motor**: **FPL, FDP to index (and middle), pronator quadratus**; **cannot make an "OK" sign (pinch sign: flat pulp-to-pulp pinch)**; **no sensory loss**. Causes: compression (same sites), **Parsonage-Turner syndrome (brachial neuritis: often the true cause: severe shoulder pain preceding weakness)**, **supracondylar fracture in children (AIN is the commonest nerve injured in extension-type fractures)**. Observe for 3-6 months (most recover); explore if no recovery.

4. Cubital tunnel syndrome (ulnar nerve at the elbow)

- **Second commonest compressive neuropathy. Sites of compression (proximal to distal):**
  - **Arcade of Struthers** (about 8 cm proximal to the medial epicondyle; medial intermuscular septum),
  - **Medial intermuscular septum**,
  - **Medial epicondyle** (subluxation, valgus deformity: **tardy ulnar nerve palsy after lateral condyle non-union with cubitus valgus**),
  - **Osborne's ligament (cubital tunnel retinaculum: roof of the cubital tunnel)** (**commonest site**),
  - **Between the two heads of FCU (Osborne's fascia)**,
  - **Deep flexor-pronator aponeurosis**.
- **Elbow flexion increases pressure and stretches the nerve** (nerve elongates about 4.5-8 mm).
- **Symptoms/signs:** paraesthesia of the **little finger and ulnar half of the ring finger** + **dorsoulnar hand (dorsal cutaneous branch: distinguishes from Guyon canal compression)**, **intrinsic weakness**, **Froment sign** (FPL substitutes for **adductor pollicis**: IPJ flexion during key pinch), **Jeanne sign** (MCPJ hyperextension of the thumb), **Wartenberg sign** (**abducted little finger: unopposed EDM (radial nerve), weak third palmar interosseous**), **claw hand (ulnar claw: ring and little fingers)**, **elbow flexion test**, Tinel.

- **Ulnar paradox: proximal (elbow) lesions cause less clawing than distal (wrist) lesions** because **FDP to ring and little fingers is also paralysed** in high lesions.
- **Treatment:** activity modification, **night splint in extension (about 45 degrees)**; surgery: **in situ decompression (open or endoscopic: first-line; equivalent outcomes to transposition, fewer complications)**, **anterior transposition (subcutaneous or submuscular)** for **unstable/subluxing nerves** or after failed decompression, **medial epicondylectomy**. **Medial antebrachial cutaneous nerve** branches at risk (painful neuroma).
- **McGowan classification** (grade I sensory, II weakness, III atrophy/paralysis): severe/long-standing disease has poorer recovery.

5. Guyon canal (ulnar tunnel) syndrome

- **Boundaries: roof: volar carpal ligament (and palmaris brevis)**, **floor: transverse carpal ligament (and pisohamate/pisometacarpal ligaments)**, **radial wall: hook of hamate**, **ulnar wall: pisiform**.
- **Zones (Gross and Gelberman):**
  - **Zone 1** (proximal to the bifurcation): **motor and sensory deficit**,
  - **Zone 2** (deep motor branch, around the hook of hamate): **pure motor**,
  - **Zone 3** (superficial sensory branch): **pure sensory**.
- **Causes: ganglion (commonest)**, **hook of hamate fracture/non-union** (golfers, baseball, cyclists' palsy), **ulnar artery thrombosis/aneurysm (hypothenar hammer syndrome)**, anomalous muscles, lipoma.
- **Dorsal cutaneous branch spared** (arises **about 5-8 cm proximal to the wrist**): **dorsal hand sensation normal** (vs cubital tunnel). **FDP to ring/little and FCU normal**.
- **Treatment:** CT/MRI to identify the cause; decompression, excision of ganglion/hamate hook.

6. Radial nerve compression

- **Radial tunnel syndrome: pain** (lateral forearm, **about 4-5 cm distal to the lateral epicondyle, over the radial tunnel**), **no motor weakness**; mimics **tennis elbow** (coexist in about 5%); **pain on resisted middle**

finger extension and resisted supination; NCS normal. **Treatment:** non-operative first; decompression.

- **Posterior interosseous nerve (PIN) syndrome: motor weakness without sensory loss**: **finger and thumb extension drop**, **wrist extends in radial deviation** (ECRL (radial nerve proper) preserved; **ECU paralysed**).
- **Compression sites in the radial tunnel (FREAS):**
  - **Fibrous bands** anterior to the radiocapitellar joint,
  - **Recurrent radial vessels ("leash of Henry")**,
  - **ECRB tendinous margin**,
  - **Arcade of Frohse (proximal edge of the supinator: commonest site)**,
  - **Supinator distal edge**.
- **Other causes: Monteggia fracture-dislocation (PIN palsy: commonest nerve injury, usually neurapraxia)**, **lipoma**, **rheumatoid synovitis of the elbow** (radiocapitellar), ganglion. **Differentiate from extensor tendon rupture** (RA): **tenodesis test** (wrist flexion extends the MCPJs if tendons are intact).
- **Wartenberg syndrome (cheiralgia paraesthetica): superficial radial nerve** compression **between brachioradialis and ECRL** (tight watch, handcuffs): dorsoradial hand paraesthesia; **Tinel**, pronation with wrist ulnar deviation provokes it.
- **Saturday night palsy** (radial nerve in the **spiral groove**): **wrist drop**, **triceps spared** (branches arise proximal).

7. Other compressive and nerve lesions

- **Thoracic outlet syndrome: neurogenic (commonest, about 95%; lower trunk C8-T1: ulnar-sided symptoms, Gilliatt-Sumner hand: wasting of the thenar eminence (APB))**, **venous (Paget-Schroetter: effort thrombosis of the subclavian vein)**, **arterial (cervical rib: subclavian artery aneurysm, emboli)**. **Causes:** cervical rib, anomalous bands, scalene hypertrophy. **Tests: Roos (elevated arm stress) test**, **Adson test**, Wright. Physiotherapy first; first rib resection/scalenectomy.
- **Suprascapular nerve:**

  - **Suprascapular notch** (under the **transverse scapular ligament**): **supraspinatus + infraspinatus** weakness,
  - **Spinoglenoid notch** (**paralabral cyst** from a **SLAP tear**): **isolated infraspinatus** weakness (painless wasting: volleyball players).
- **Quadrilateral space syndrome: axillary nerve + posterior circumflex humeral artery** (fibrous bands; throwers): posterior shoulder pain, **teres minor** atrophy on MRI.
- **Long thoracic nerve palsy: medial scapular winging** (serratus anterior). **Spinal accessory nerve palsy** (posterior triangle lymph node biopsy): **lateral winging** (trapezius) with shoulder drooping.
- **Double-crush syndrome:** proximal compression (cervical radiculopathy) makes the distal nerve more susceptible to compression.

8. Tendon transfers (for nerve palsies)

- **Principles: adequate strength (lose one MRC grade after transfer: muscle of grade 5 needed)**, **expendable donor**, **similar excursion** (**wrist flexors about 33 mm, finger extensors about 50 mm, finger flexors about 70 mm**: **"3-5-7" rule**), **straight line of pull**, **one tendon one function**, **synergistic transfers easier to retrain** (wrist flexion with finger extension), **supple joints, stable soft-tissue bed**, **tissue equilibrium** (wait for swelling/scar to settle).
- **Radial nerve palsy (high): pronator teres to ECRB** (wrist extension), **FCR or FCU to EDC** (finger extension), **palmaris longus to EPL (rerouted)** (thumb extension). **Early "internal splint" (PT to ECRB)** while awaiting nerve recovery.
- **Low median nerve palsy (opposition): opponensplasty**: **EIP (Burkhalter)**, **FDS ring (Bunnell)**, **palmaris longus (Camitz: for severe CTS; mainly abduction)**, **abductor digiti minimi (Huber: congenital thumb hypoplasia)**.
- **High median:** + **ECRL to FDP index/middle**, **brachioradialis to FPL**.
- **Ulnar nerve palsy: clawing correction**: **Zancolli capsulodesis** (MCPJ volar plate), **Stiles-Bunnell (FDS lasso)**, **ECRL/ECRB transfer (Brand) to the lateral bands**; **key pinch**: **ECRB to adductor pollicis (Smith)**, **FDS ring to adductor**; **index abduction**: **EIP or APL slip to first dorsal**

interosseous; **Bouvier manoeuvre** (blocking MCPJ hyperextension corrects the claw: **intact extensor mechanism**: **simple (static) procedure suffices** if positive).

Exam pearl: **Seddon/Sunderland: neurapraxia (I) no Tinel, full recovery; axonotmesis II full, III/IV incomplete/none; neurotmesis V needs surgery. Regeneration 1 mm/day. Motor end plates lost after 12-18 months: nerve transfers. EMG at 3 weeks. Preserved SNAP with anaesthetic skin = preganglionic.**

Exam pearl: **CTS: palm spared (palmar cutaneous branch); Durkan most sensitive; Semmes-Weinstein for early compression; incomplete release = commonest cause of persistent symptoms; transligamentous motor branch at risk. Pronator syndrome: palm involved, commonest between pronator teres heads. AIN: no OK sign, no sensory loss; consider Parsonage-Turner.**

Exam pearl: **Cubital tunnel: dorsal ulnar sensation lost (vs Guyon: spared); Osborne's ligament commonest; in situ decompression first-line. Ulnar paradox: distal lesions claw more. Guyon zone 2 = pure motor (hook of hamate). PIN palsy: radial deviation of the wrist on extension; arcade of Frohse; RA: tenodesis test excludes tendon rupture.**

Exam pearl: **Tendon transfers: lose one grade; 3-5-7 cm excursion; radial palsy: PT to ECRB, FCR/FCU to EDC, PL to EPL. Opponensplasty: EIP/FDS ring/Camitz (PL)/Huber (ADM).**

Brachial plexus

1. Anatomy

- **Roots C5-T1** (variations: **prefixed C4-C8**, **postfixed C6-T2**). **Mnemonic: Roots, Trunks, Divisions, Cords, Branches** ("Randy Travis Drinks Cold Beer").
- **Roots** (between scalenus anterior and medius): branches **dorsal scapular nerve (C5: rhomboids, levator scapulae)**, **long thoracic nerve (C5-7: serratus anterior)**, contribution to the phrenic nerve (C5).
- **Trunks** (posterior triangle of the neck): **upper (C5-6)**, **middle (C7)**, **lower (C8-T1)**. Upper trunk branches: **suprascapular nerve (supraspinatus, infraspinatus)**, **nerve to subclavius**.
- **Divisions** (behind the clavicle): anterior and posterior; **no branches**.
- **Cords** (named by relation to the **axillary artery**, second part, deep to pectoralis minor):
  - **Lateral (C5-7): lateral pectoral nerve**, **musculocutaneous nerve**, lateral root of the median nerve.
  - **Medial (C8-T1): medial pectoral nerve**, **medial cutaneous nerves of arm and forearm**, **ulnar nerve**, medial root of the median nerve.
  - **Posterior (C5-T1): upper and lower subscapular nerves** (subscapularis, teres major), **thoracodorsal nerve (C6-8: latissimus dorsi)**, **axillary nerve (C5-6)**, **radial nerve (C5-T1)**.
- **Erb's point:** junction of C5 and C6 roots (upper trunk), **about 2-3 cm above the clavicle** behind the sternocleidomastoid.
- **Sympathetic chain: T1 contributes to the stellate ganglion: Horner syndrome (ptosis, miosis, anhidrosis, enophthalmos)** = **T1 preganglionic avulsion**.

2. Levels of injury: preganglionic vs postganglionic

Feature	Preganglionic (root avulsion; proximal to the dorsal root ganglion)	Postganglionic (distal to the DRG)
Recovery potential	**None (no spontaneous recovery, cannot be repaired directly)**	**Possible** (repair/graft, spontaneous)
Horner syndrome	**Present (T1)**	Absent
Proximal muscle weakness: **serratus anterior (long thoracic), rhomboids (dorsal scapular)**, **paraspinals**	**Paralysed** (winging)	**Spared**
Hemidiaphragm paralysis (phrenic, C5)	Possible	Absent
Tinel sign in the neck (supraclavicular)	**Absent**	**Present**
Severe neuropathic pain (burning, deafferentation)	**Common**	Less
SNAP (sensory nerve action potential)	**Preserved (despite anaesthetic skin)** (DRG cell bodies intact, peripheral axon intact)	**Absent**
Imaging (CT myelography/MRI)	**Pseudomeningocele**, absent rootlets	Normal roots
Histamine test (triple response)	Preserved	Absent
Fractures	Transverse process fractures of the cervical vertebrae, first rib fracture	Clavicle fracture

- **C5-C6 are most often postganglionic ruptures** (strong ligamentous attachment (transverse radicular ligaments) in the gutter of the transverse

processes); **C8-T1 are most often avulsed (preganglionic)** (less protected).

- **Avulsion: central (intradural) avulsion** of rootlets; **peripheral avulsion** at the foramen.

3. Adult traumatic brachial plexus injury

- **Mechanism: high-energy traction**: **motorcycle accidents (young men)**: **head/neck forced away from a depressed shoulder** (upper plexus: C5-6) or **arm forced into abduction** (lower plexus: C8-T1).
- **Patterns:**
  - **Upper (C5-6) (Erb):** shoulder abduction/external rotation and elbow flexion lost; **"waiter's tip" posture**.
  - **Extended upper (C5-7):** + wrist and finger extension weakness.
  - **Lower (C8-T1) (Klumpke):** intrinsic hand paralysis, claw hand, ± **Horner** (rare in isolation in adults).
  - **Pan-plexus (C5-T1)**: commonest pattern in adults (**about 50-75%**): flail anaesthetic arm.
- **Associated injuries: subclavian/axillary artery injury**, **scapulothoracic dissociation** (**lateral displacement of the scapula on chest radiograph**, plexus avulsion + vascular injury: **poor prognosis**), **clavicle fracture**, **first rib fracture**, **head and cervical spine injuries**, **shoulder dislocation (infraclavicular plexus and axillary nerve)**.
- **Infraclavicular injuries** (cords/branches: shoulder dislocation, proximal humerus fractures) have **better prognosis** (postganglionic).
- **Investigations:**
  - **Chest radiograph** (hemidiaphragm, rib fractures), **cervical spine**.
  - **MRI** or **CT myelography** (**pseudomeningoceles**: avulsion; CT myelography more sensitive for rootlet avulsion).
  - **NCS/EMG at about 3-4 weeks** (SNAPs, paraspinal denervation: preganglionic).
  - **Intraoperative** nerve stimulation, **somatosensory evoked potentials (SSEP)**, frozen section (ganglion cells: avulsion).

Management

- **Timing:**
  - **Open/sharp injuries: immediate exploration and repair**.
  - **Closed traction injuries with suspected avulsion (Horner sign, pseudomeningoceles): early exploration (within about 3 months)** for nerve transfers/grafts.
  - **Closed injuries with likely postganglionic lesions: observe 3 months**; explore if no clinical/EMG recovery (**ideally within 3-6 months**; **results deteriorate after about 6 months**; **motor end-plate degeneration by 12-18 months**).
  - **Vascular injury:** emergency repair (± plexus exploration).
- **Surgical options:**
  - **Neurolysis** (neuroma-in-continuity with conduction across it).
  - **Nerve grafting (sural nerve)** from viable roots (postganglionic).
  - **Nerve transfers (neurotisation):**
  - **Elbow flexion: Oberlin transfer** (**fascicle of the ulnar nerve to the biceps motor branch**), **double fascicular transfer** (ulnar fascicle to biceps + **median fascicle to brachialis**) (Mackinnon), **intercostal nerves to musculocutaneous nerve**, **spinal accessory to musculocutaneous**, **phrenic nerve**, **contralateral C7**.
  - **Shoulder: spinal accessory (distal) to suprascapular nerve** (via posterior approach preferred), **radial nerve (triceps long-head branch) to axillary nerve (Leechavengvongs/Somsak transfer)**.
  - **Priority in pan-plexus injury: elbow flexion first**, then **shoulder stability/abduction**, then hand sensation, then wrist/finger function.
  - **Free functioning muscle transfer (gracilis)** reinnervated by spinal accessory or intercostal nerves for late presentation or pan-plexus (elbow flexion and/or finger flexion: **Doi double free muscle**).
  - **Tendon transfers** (incomplete injuries, late), **Steindler flexorplasty** (proximal advancement of the flexor-pronator origin for elbow flexion), **latissimus/pectoralis transfer for elbow flexion**, **trapezius transfer for shoulder abduction**.
  - **Shoulder arthrodesis** (flail shoulder with **intact scapulothoracic muscles: trapezius, serratus anterior**): **about 20-30 degrees**

abduction, 20-30 degrees flexion, 40 degrees internal rotation ("salute position"/hand-to-mouth).
    - **Wrist arthrodesis**, **amputation** (above elbow with shoulder fusion: seldom improves function; **does not relieve pain**).
- **Neuropathic (deafferentation) pain:** pain medication (gabapentinoids), **dorsal root entry zone (DREZ) lesioning** for intractable pain in avulsions.

4. Obstetric (birth) brachial plexus palsy (BBPP/NBPP)

- **Incidence about 1-2 per 1,000 live births. Risk factors: shoulder dystocia (strongest)**, **macrosomia (large for gestational age, maternal diabetes)**, **breech delivery** (**bilateral, upper-root avulsions**), instrumental delivery, prolonged labour, previous affected child.
- **Narakas classification:**
    - **Group I: C5-6** (**Erb's palsy**; commonest; **best prognosis: about 90% spontaneous recovery**): **waiter's tip** (adducted, internally rotated shoulder, extended elbow, pronated forearm, flexed wrist).
    - **Group II: C5-7** (extended Erb): + wrist/finger extension loss (wrist drop).
    - **Group III: C5-T1 without Horner** (total palsy, flail arm).
    - **Group IV: C5-T1 with Horner syndrome** (**worst prognosis**).
    - (**Klumpke palsy, isolated C8-T1, is very rare.**)
- **Differential diagnosis (pseudoparalysis): clavicle fracture**, proximal humerus physeal fracture, **septic arthritis/osteomyelitis of the shoulder**, cerebral palsy.
- **Prognosis: recovery of biceps (elbow flexion) by 3 months** is the classic prognostic indicator (Gilbert and Tassin): **absence of biceps function at 3 months suggests poor spontaneous recovery: consider microsurgery (3-9 months)**. **Toronto Test Score** (elbow flexion, elbow extension, wrist, finger and thumb extension at 3 months; **score under 3.5 = surgery**), **cookie test at 9 months** (hand-to-mouth with the arm adducted). **Horner syndrome and total palsy: earlier surgery (about 3 months)**.
- **Management:**
    - **Physiotherapy (passive ROM: shoulder external rotation, abduction; prevent contracture)** from about 2-3 weeks.

  - **Primary microsurgical reconstruction** (neuroma excision + sural grafts, nerve transfers) at **3-9 months** if no biceps recovery.
  - **Assessment scales: Active Movement Scale (AMS)**, **Mallet classification** (shoulder function: abduction, external rotation, hand to neck, hand to back, hand to mouth; grades I-V).
- **Secondary deformities:**
  - **Internal rotation contracture of the shoulder** (**subscapularis/pectoralis major contracture, weak external rotators**): **glenohumeral dysplasia (posterior subluxation, glenoid retroversion, pseudoglenoid formation)**: **Waters classification (types I-VII)** on MRI/ultrasound. Monitor **passive external rotation** (in adduction); **ultrasound screening** in infancy.
  - **Treatment: botulinum toxin to subscapularis/pectoralis**, **open/arthroscopic release + reduction (subscapularis lengthening/release)**, **tendon transfer of latissimus dorsi and teres major to the rotator cuff (L'Episcopo/Hoffer/Sever)** for external rotation and abduction, **humeral external rotation (derotation) osteotomy** (late, with severe glenoid deformity).
  - **Elbow flexion contracture**, **radial head dislocation (posterior)**, **forearm supination deformity** (biceps rerouting, forearm osteotomy), **"trumpet sign"** (shoulder abducts on attempted hand-to-mouth: weak external rotation).

5. Other brachial plexus conditions

- **Parsonage-Turner syndrome (neuralgic amyotrophy, acute brachial neuritis): sudden severe shoulder pain** (often after infection, vaccination, surgery) followed days later by **patchy weakness** (**long thoracic, suprascapular, AIN, axillary nerves** commonly); **pain resolves before weakness**; **hourglass constrictions** of nerve fascicles on high-resolution ultrasound/MRI. **Self-limiting** (recovery over months to years); steroids early; surgery (interfascicular neurolysis) for hourglass constrictions without recovery.
- **Radiation-induced plexopathy** (breast cancer; **myokymia** on EMG; usually **upper plexus**, painless) vs **tumour infiltration (Pancoast: lower**

plexus, painful, Horner).

- **Thoracic outlet syndrome** (see nerve compression notes).
- **Stingers/burners** (transient upper trunk traction or root compression in contact sport): unilateral burning dysaesthesia, transient weakness; return to play once symptoms resolve fully; recurrent: investigate (stenosis).
- **Brachial plexus tumours: schwannoma (commonest)**, **neurofibroma** (NF1, plexiform), **malignant peripheral nerve sheath tumour** (NF1; rapid growth, pain).

6. Examination checklist (rapid)

- **Proximal (preganglionic) clues: rhomboids** (dorsal scapular, C5), **serratus anterior** (long thoracic, C5-7: winging), **paraspinals** (posterior primary rami), **diaphragm** (phrenic), **Horner** (T1).
- **Trunk level: supraspinatus/infraspinatus** (suprascapular nerve: from the upper trunk).
- **Cord level: pectoralis major** (clavicular head: lateral cord (lateral pectoral); sternal head: medial cord (medial pectoral)), **latissimus dorsi** (thoracodorsal: posterior cord).
- **Terminal branches:** axillary (deltoid, regimental badge sensation), musculocutaneous (biceps, lateral forearm sensation), radial, median, ulnar.

> Exam pearl: **Preganglionic avulsion signs: Horner (T1), winging (long thoracic), rhomboid paralysis (dorsal scapular), paraspinal denervation, hemidiaphragm, absent Tinel, preserved SNAP with anaesthetic skin, pseudomeningocele. C5-6 rupture (postganglionic); C8-T1 avulse.**

> Exam pearl: **Adult closed plexus injury: explore by 3-6 months; motor end plates lost by 12-18 months. Elbow flexion priority: Oberlin (ulnar fascicle to biceps branch); shoulder: spinal accessory to suprascapular + triceps branch to axillary. Late: free functioning gracilis; flail shoulder: arthrodesis.**

Exam pearl: **Obstetric palsy: shoulder dystocia; Narakas I (C5-6) best, IV (with Horner) worst; no biceps recovery at 3 months (Gilbert) or Toronto score under 3.5 = microsurgery at 3-9 months. Secondary internal rotation contracture: glenoid dysplasia (Waters); latissimus/teres major transfer (L'Episcopo/Hoffer).**